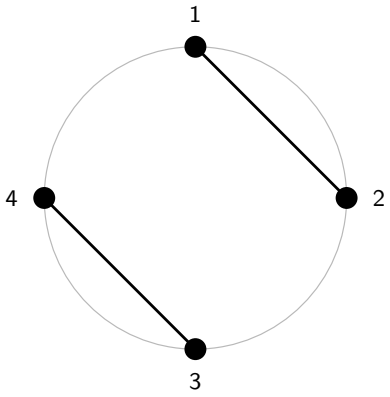
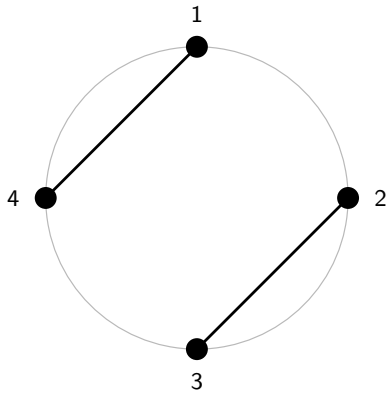
Week 12 / Catalan bijections / K–1

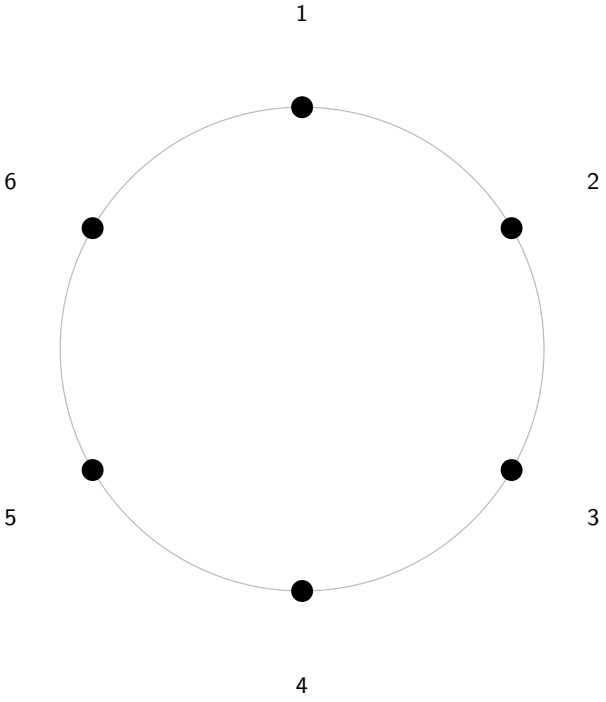
Use each dot once. Strings stay inside the circle and do not cross. A dot with a different partner makes a different pairing.

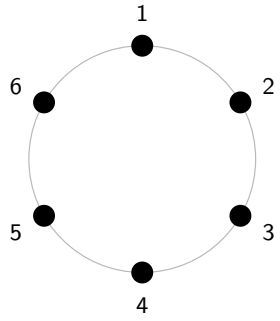
The two pairings of four dots:

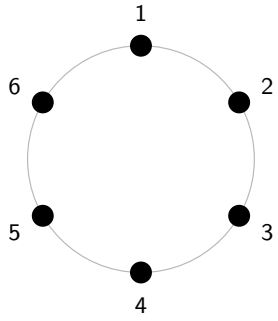


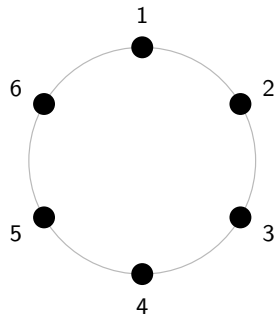


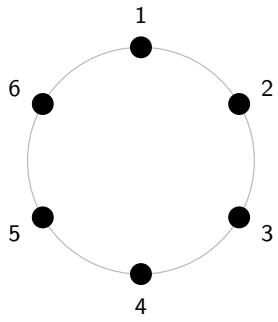
Problem 1: Find every way to pair six dots.

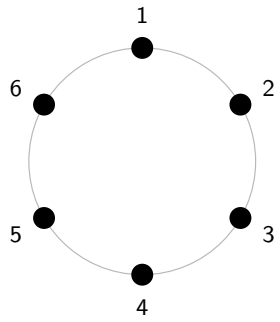


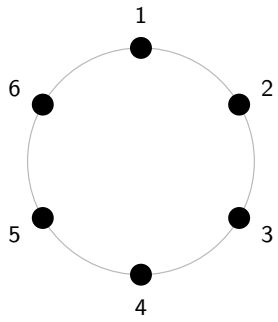




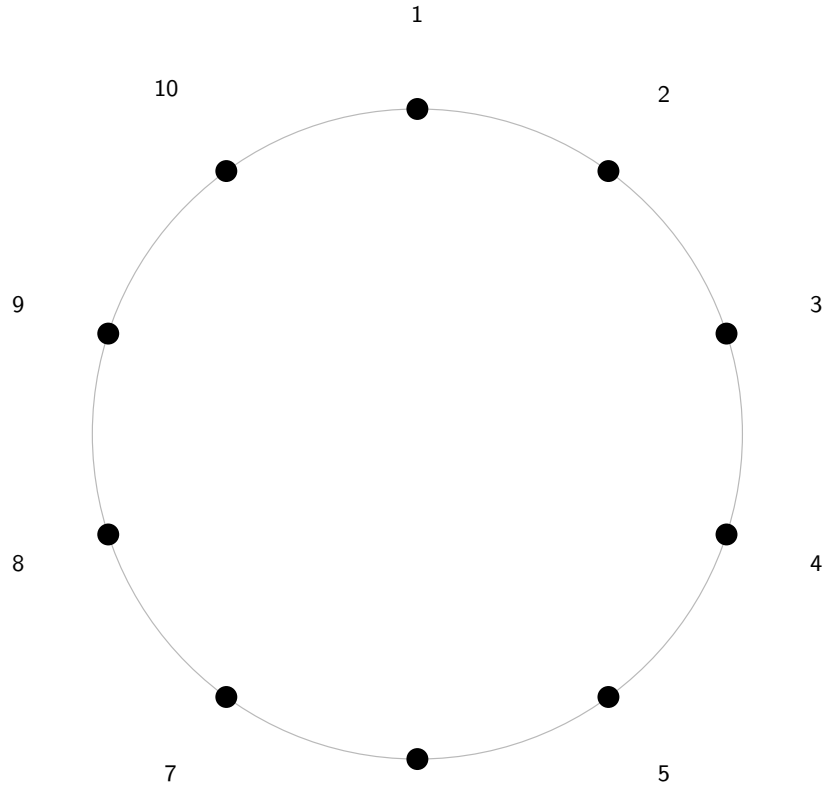


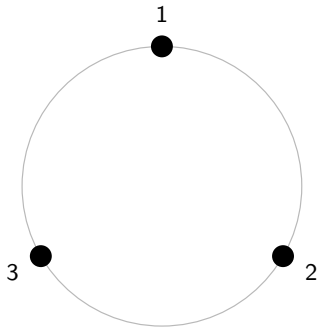


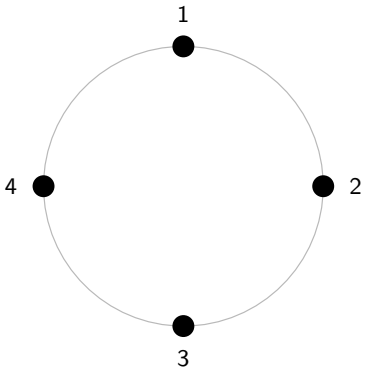


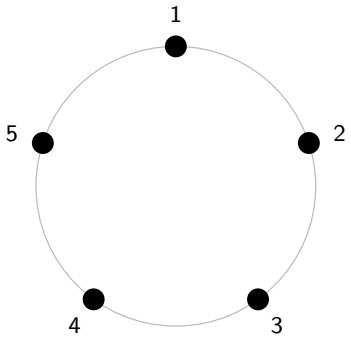


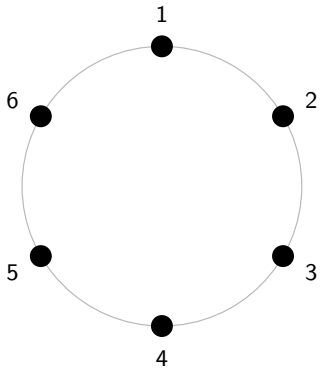
Problem 2: Pair all the dots in each circle if you can. If you cannot, show what gets in the way.

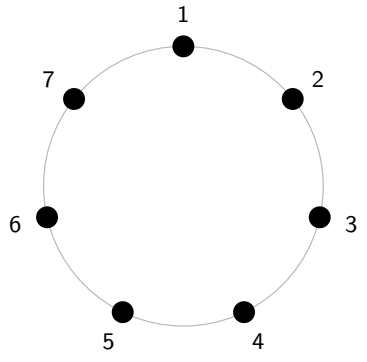




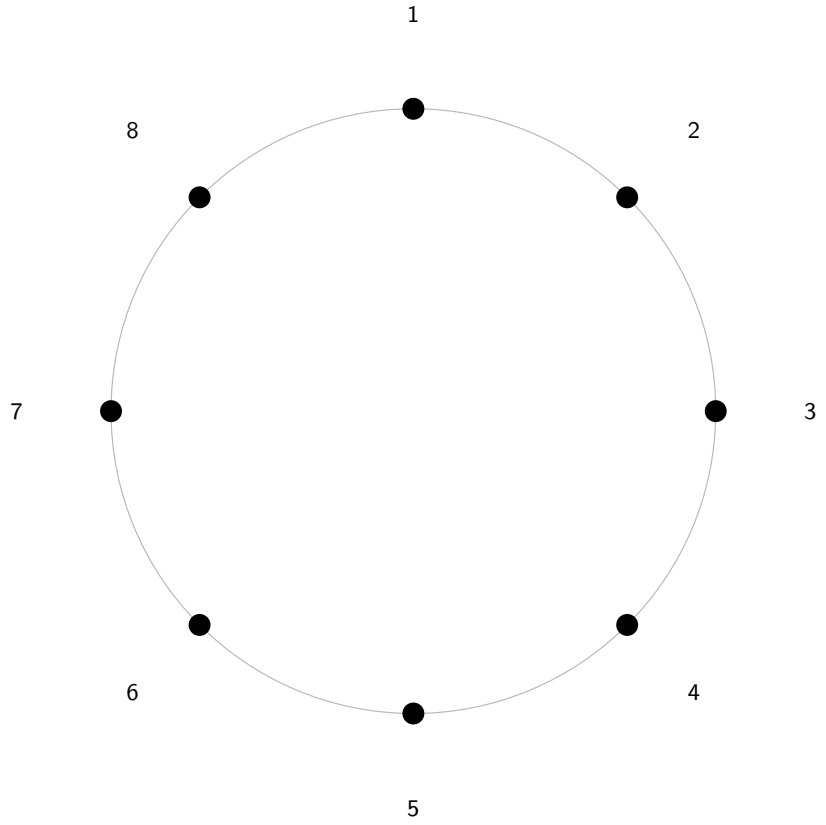


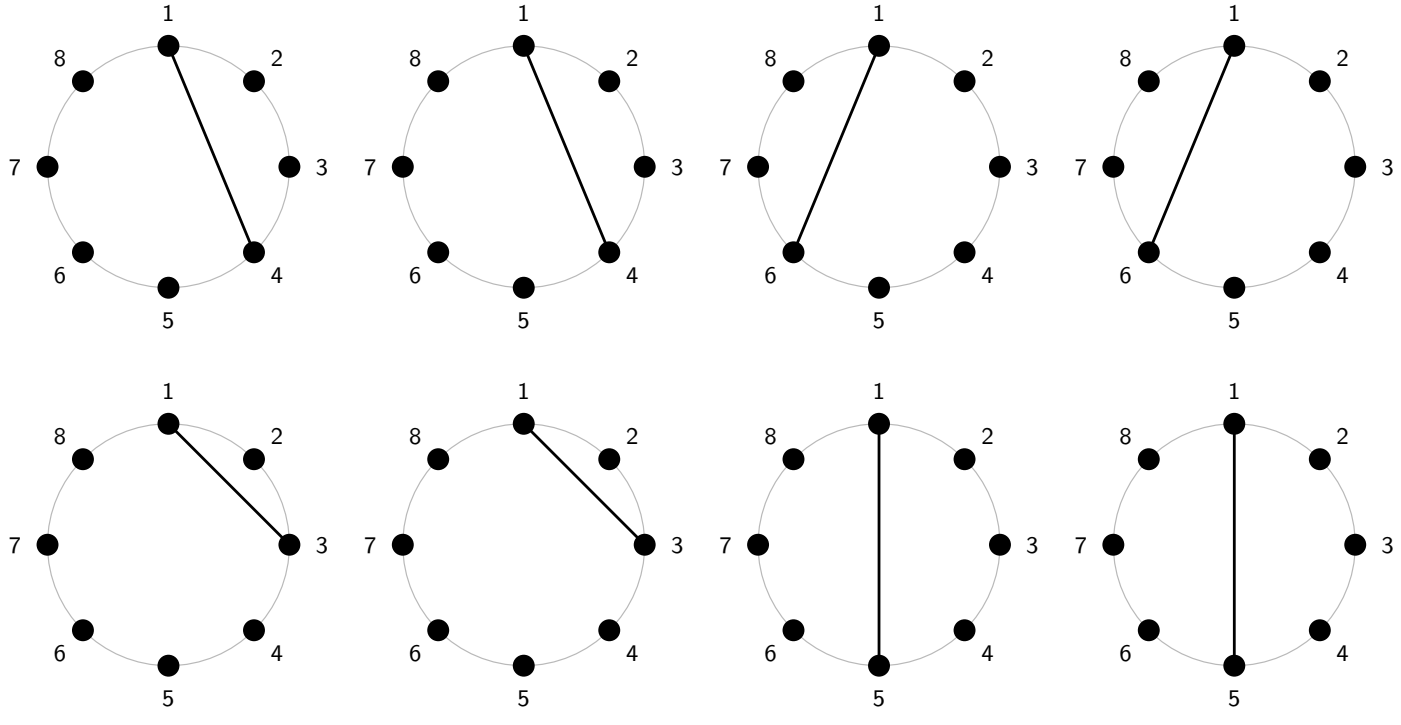




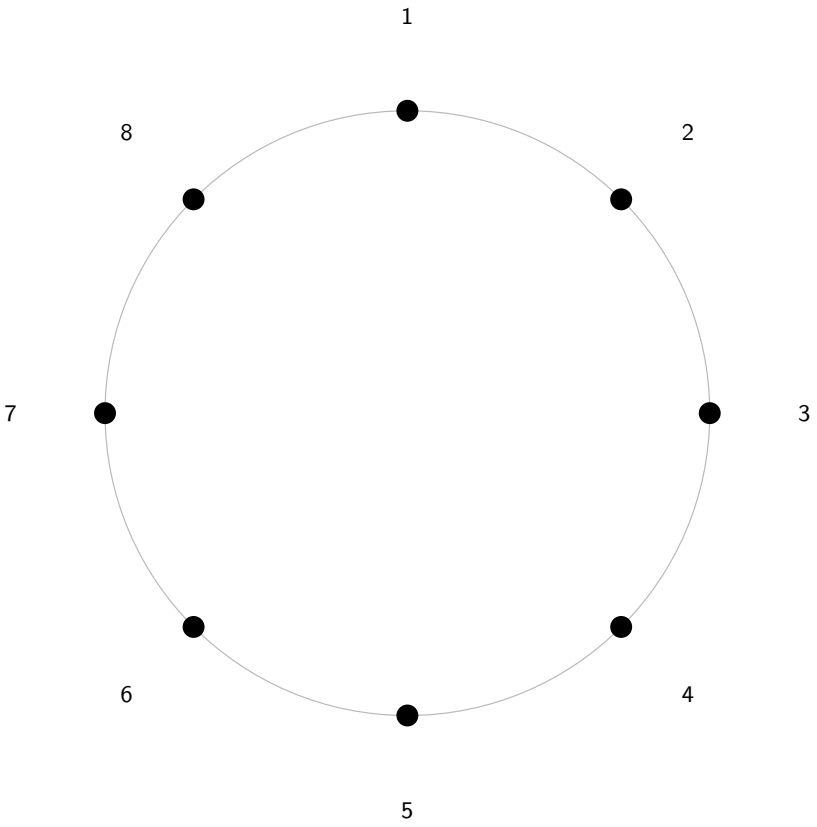


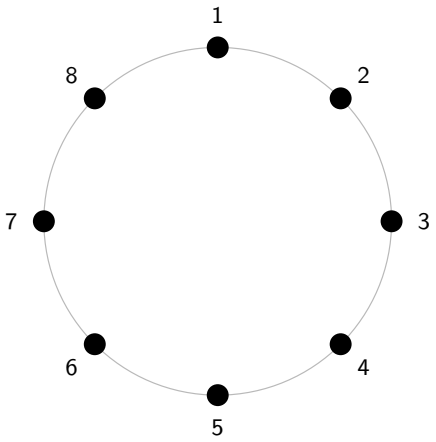
Problem 3: For each drawn pair, put a string on the large circle and leave it there. Find every way to finish the pairing, or show why you cannot.

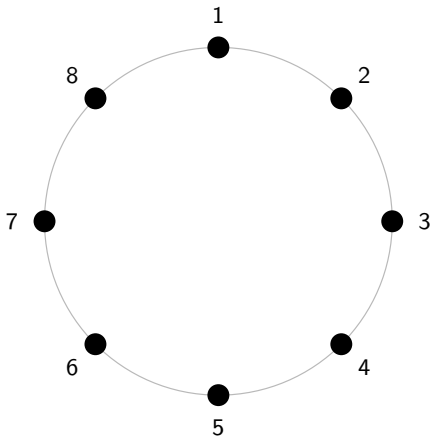


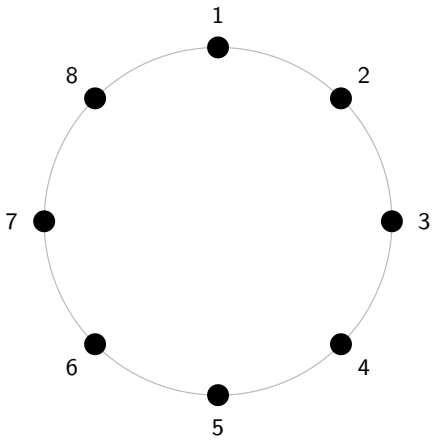


Problem 4: Pair eight dots in as many different ways as you can.

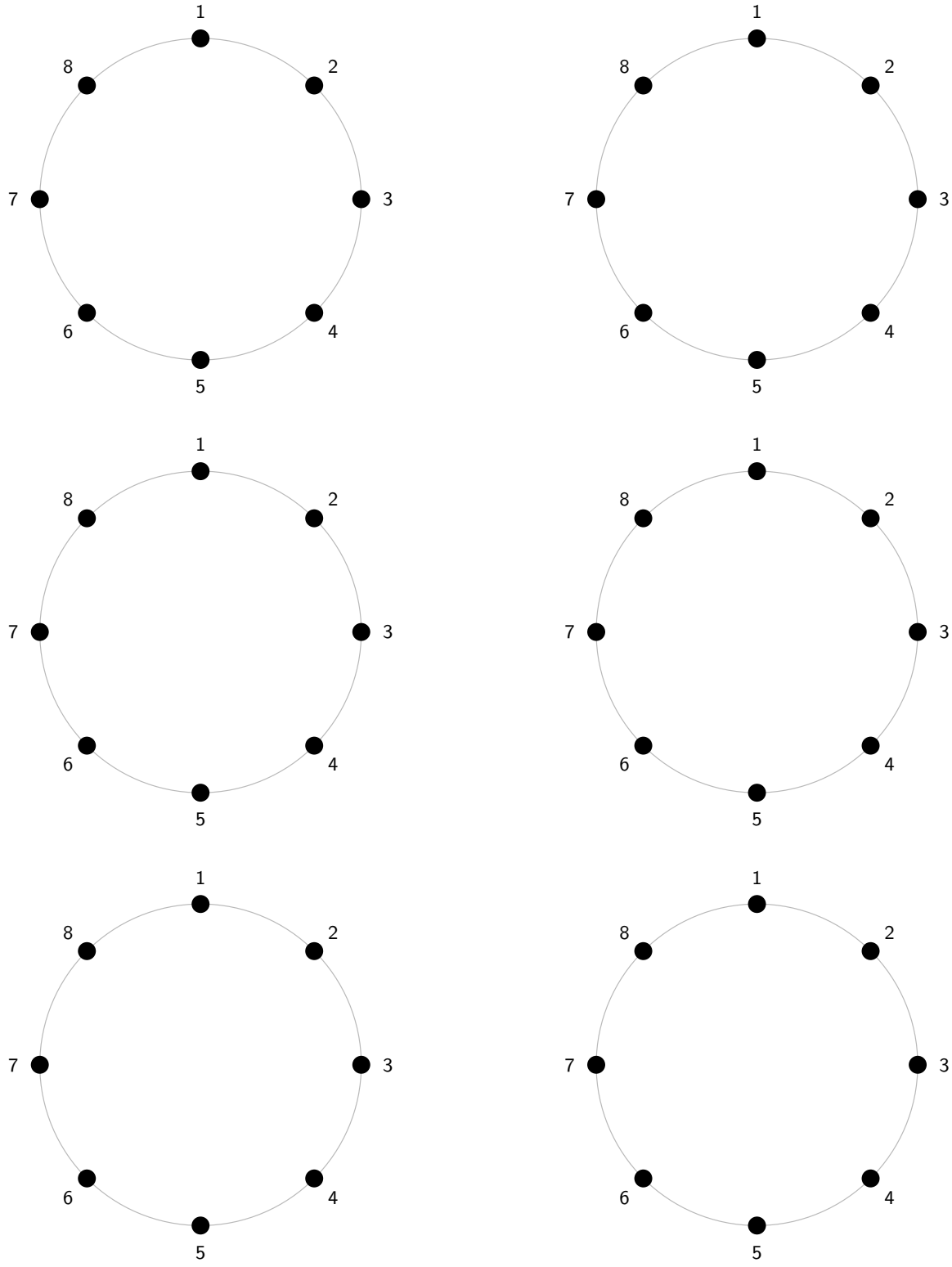




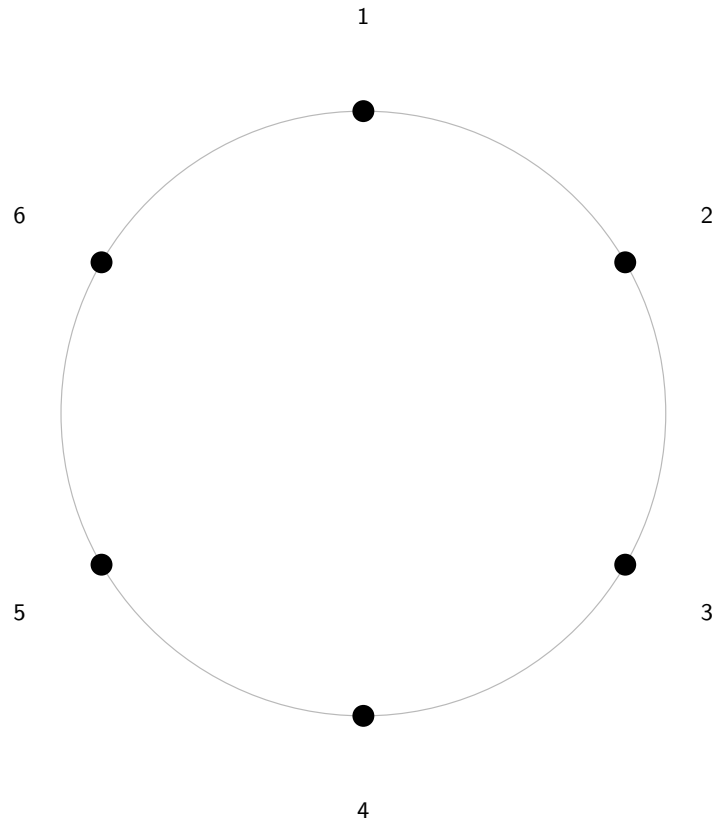




Problem 4 (continued)



Problem 5: Check your six-dot pairings from Problem 1. Show how you know that no pairing is missing.



Problem 6: Can you pair eight dots without joining any two neighbors around the circle? Show why your answer is right.

